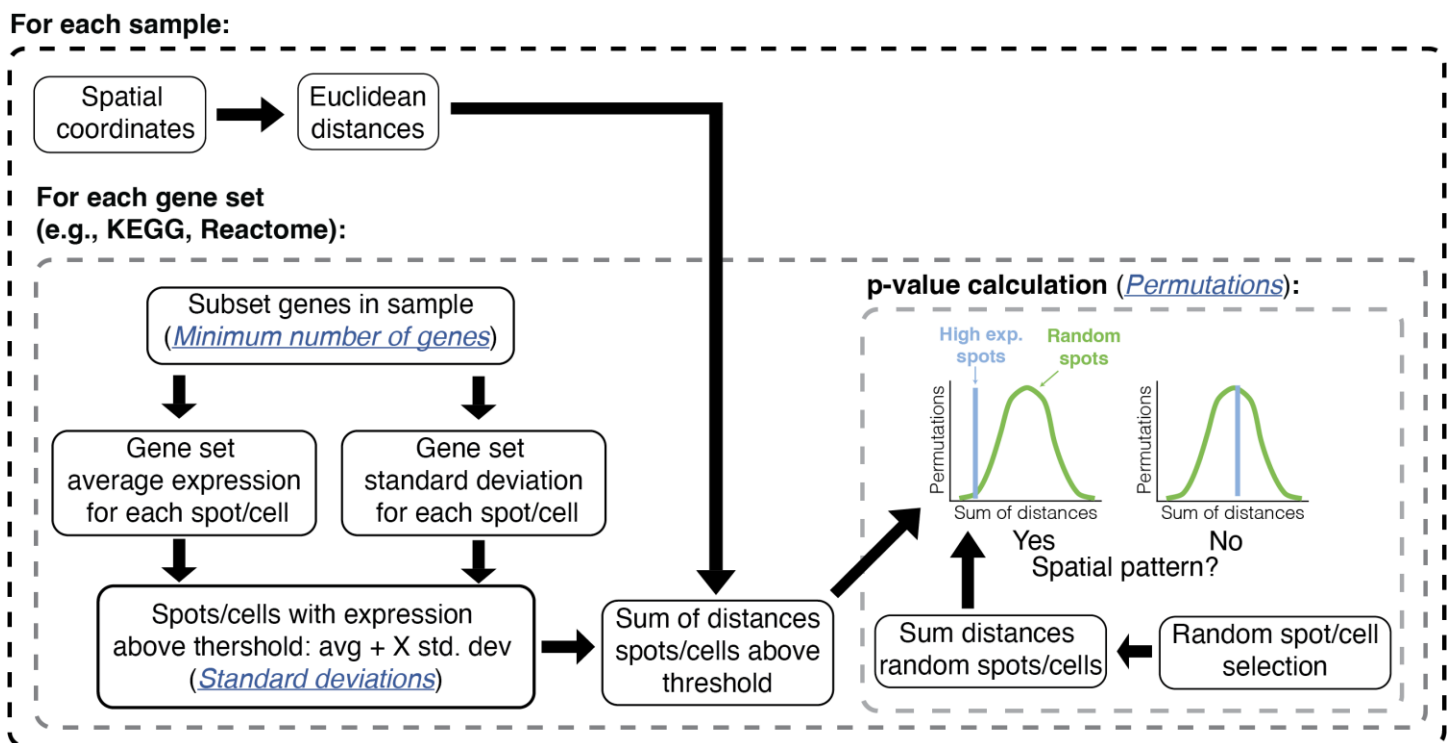


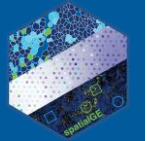
In gene expression analysis, researchers study changes not only at the gene level but also at the pathway (or gene set) level. Traditionally, this is done through gene set enrichment analysis (GSEA). When analyzing spatial gene expression, researchers may also want to identify gene sets that are "spatially enriched." In spatialGE, a spatially enriched gene set is one where spots or cells with high expression of the gene set are spatially aggregated (i.e., close one to the other). For example, in many tumor tissues, certain gene sets related to the cell cycle might be highly expressed in tumor areas, creating a pattern of spatial aggregation.

The **Spatial gene set enrichment** module in spatialGE performs this analysis. It utilizes the STenrich function from the R package spatialGE, which is a modified version of the method proposed by [Hunter et al. \(2021\)](#). The diagram below illustrates the STenrich algorithm:



In **Spatial gene set enrichment**, each sample's spatial enrichment is analyzed separately. First, Euclidean distances are calculated among all spots/cells. Then, the gene expression of the sample is subset to the gene set being tested. If too few genes remain after subsetting (based on the **Minimum number of genes** parameter), the gene set is omitted for that sample. The average expression and standard deviation of those genes are calculated for each spot/cell. Next, spots/cells with gene set expression above the average gene set expression across all spots/cells are identified. The threshold for defining these high gene set expression spots/cells is set by the average gene set expression plus a specified number of standard deviations (**Standard deviations** parameter). Finally, the sum of the Euclidean distances between the high-expression spots/cells is calculated.

The next step involves a permutation process to generate a null distribution, testing if the sum of distances among high expression spots is smaller than expected. A random sample of spots/cells, regardless of expression, is



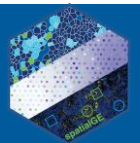
selected, matching the number of high-expression spots/cells. The sum of distances among these randomly sampled spots/cells is then calculated. This random selection is repeated as many times as specified by the **Permutations** parameter. Finally, a p-value is calculated by counting how often the sum of random distances exceeds the sum of distances among high-expression ROIs/spots/cells. If the sum of random distances is higher most of the time, the null hypothesis of no spatial aggregation is rejected, indicating that spots/cells with high gene set expression are more aggregated than expected by chance.

When conducting an analysis with the **Spatial gene set enrichment** module, consider the following important notes:

- Two enrichment metrics per spot/cell are available: 1) Gene set enrichment score (provided via the “gsva” R package) and 2) the average expression of the genes in a set. Using a gene set enrichment score is generally favored over the average expression.
- Reference Genome: Pay special attention to the reference genome used for annotating gene counts. Annotation is one of the first steps in spatial transcriptomics analysis, and it is performed before input to spatialGE. If transcripts are annotated with a mouse genome (or another species), use the appropriate gene set database, as gene names may not match a human gene set database (for example due to lower vs. upper case names). Additionally, gene homology issues might arise.
- Permutations: More permutations increase execution time but result in more accurate p-value estimates.
- Standard deviations: Increasing the number of standard deviations results in fewer spots/cells being selected as “high expression” (stringent analysis). However, this makes it more challenging to detect gene sets showing spatial aggregation. Values between 1 to 2 standard deviations are likely suitable for most studies.
- Minimum number of spots (or cells): Adjusting this parameter allows control over the size of the structures studied. For example, small immune infiltrates can be detected with a smaller minimum number of spots, while larger tissue domain-level differences require a larger minimum number of spots/cells.
- Random seed number: Note the random seed number used for the test (Seed number parameter) to reproduce the same results later. Running the analysis multiple times with different seed numbers is advisable to check for consistency.

With the algorithm explained, begin an analysis by selecting a gene set in the **Input gene sets** drop-down. For this example, select the “KEGG – human” option (see also note above regarding selection of database and genome used in annotation). Additional functionality will be included to allow for custom gene sets to be entered. Currently, the Average and GSEA score checkboxes are disabled, and only the average gene set expression is calculated.

To test for spatial enrichment, select a built-in gene set from the **Input gene sets** drop-down menu. Alternatively, users can enter custom gene sets (“UPLOAD you own gene database” option). The spatialGE app takes gene sets in the [GMT file format](#). In this format, each row is a gene set. The first column contains gene set names, the second column can contain any type of comments. Third and subsequent column contain genes for the gene set in that row. The GMT file does not include headers:



|    | A                                        | B            | C        | D        | E        | F       | G        | H        | I      |
|----|------------------------------------------|--------------|----------|----------|----------|---------|----------|----------|--------|
| 1  | CITRATE_CYCLE_TCA_CYCLE                  | test_comment | IDH3B    | DLST     | PCK2     | CS      | PDHB     | PCK1     | PDHA1  |
| 2  | PENTOSE_PHOSPHATE_PATHWAY                | test_comment | RPE      | RPIA     | PGM2     | PGLS    | PRPS2    |          |        |
| 3  | PENTOSE_AND_GLUCURONATE_INTERCONVERSIONS | test_comment | UGT1A10  | UGT1A8   | RPE      | UGT1A7  | UGT1A6   | UGT2B28  | UGT1A5 |
| 4  | GLUTATHIONE_METABOLISM                   | test_comment | SRM      | GGT1     | GSTP1    | GSTT2   | GSTT1    | GSTZ1    | RRM2B  |
| 5  | ASTHMA                                   | test_comment | HLA-DRB4 | HLA-DRB5 | HLA-DOA  | HLA-DOB |          |          |        |
| 6  | AUTOIMMUNE_THYROID_DISEASE               | test_comment | HLA-DOA  | HLA-DOB  | CD80     | CD86    | CD28     | IFNA5    | IFNA4  |
| 7  | ALLOGRAFT_REJECTION                      | test_comment | HLA-DRB4 | HLA-DRB5 | HLA-DOA  | HLA-DOB | HLA-DRB3 | CD80     | CD86   |
| 8  | GRAFT_VERSUS_HOST_DISEASE                | test_comment | HLA-DRB4 | KIR2DL1  | HLA-DRB5 | HLA-DOA | HLA-DOB  | HLA-DRB3 | CD80   |
| 9  | VIRAL_MYOCARDITIS                        | test_comment | CASP9    | CASP8    | HLA-DOA  | HLA-DOB |          |          |        |
| 10 |                                          |              |          |          |          |         |          |          |        |
| 11 |                                          |              |          |          |          |         |          |          |        |

**Note:** When creating GMT files using Excel, remember to save the file as a tab-delimited file.

For this example, choose the "HALLMARK – human" option (refer to the note above regarding the selection of the database and genome used in annotation). The “GSEA score” selection will be kept for this tutorial.

**spatialGE**

- About spatialGE
- How to get started
- FAQ
- Contact us
- My Projects
- TESTPROJECT
- Import data
- QC & data transformation
- Visualization
- Spatially variable genes
- Spatial gene set enrichment**
- Spatial domain detection

**Spatial Gene Set Enrichment**  
Documentation

Spatial gene set enrichment

Spatial gene set enrichment with STenrich

Detect genes showing spatial expression patterns (e.g., hotspots). This method tests if spots/cells with high average expression (or enrichment score) of a gene set, shows evidence of spatial aggregation. High expression/score spots or cells are identified using a threshold (average expression/score + X standard deviations). Currently, the expression of a gene set is calculated as the average of the genes within a set. In the future, an option to run STenrich on enrichment scores will be provided.

Select/upload a gene set database

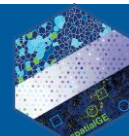
HALLMARK - human

☒ GSEA score ☐ Average

Permutations: 100

Seed number (permutation): 12345

Set the number of permutations to 1000 by entering this value in the **Permutations** textbox. Leave the other parameters unchanged, but feel free to experiment with different combinations to check for changes in the detected gene sets. Finally, click the **RUN STENRICH** button. The process may take some time, so it's recommended to select the radio button for email notification, which appears after the **RUN STENRICH** button is clicked.



FAQ

Contact us

My Projects

TESTPROJECT

Import data

QC & data transformation

Visualization

Spatially variable genes

**Spatial gene set enrichment**

Spatial domain detection

Phenotyping

Select/upload a gene set database

HALLMARK - human

☒ GSEA score ☐ Average

Permutations: 1000

Seed number (permutation): 12345

Minimum number of spots: 5

Minimum number of genes: 5

Standard deviations: 1

**RUN STENRICH**

[Download parameter log](#)

[Download zipped results](#)

After the process is completed, the results are displayed as both a table (blue arrow) and a heatmap (orange arrow). The table includes filtering functionality, allowing users to search for specific gene sets (indicated by the red arrow). It also supports sorting (indicated by the green arrow), which can be useful for arranging results in decreasing order of p-values, for example.

spatialGE

About spatialGE

How to get started

FAQ

Contact us

My Projects

TESTPROJECT

Import data

QC & data transformation

Visualization

Spatially variable genes

**Spatial gene set enrichment**

Spatial domain detection

DOWNLOAD RESULTS (EXCEL)

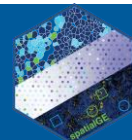
Summary Results by sample

ductal\_carcinoma breast\_cancer\_section1 breast\_cancer\_section2 breast\_cancer\_wta

CONTINUOUS SCROLLING

| Gene set          | Genes in sample | Genes in set | Proportion of gene... | p-value | Adjusted p-value |
|-------------------|-----------------|--------------|-----------------------|---------|------------------|
| HALLMARK_HYPO...  | 84              | 200          | 0.42                  | 0       | 0                |
| HALLMARK_ADIP...  | 97              | 200          | 0.485                 | 0       | 0                |
| HALLMARK_ESTR...  | 63              | 200          | 0.315                 | 0       | 0                |
| HALLMARK_ESTR...  | 70              | 200          | 0.35                  | 0       | 0                |
| HALLMARK_ANDR...  | 58              | 101          | 0.574                 | 0       | 0                |
| HALLMARK_PROT...  | 63              | 96           | 0.656                 | 0       | 0                |
| HALLMARK_INTER... | 75              | 97           | 0.773                 | 0       | 0                |
| HALLMARK_INTER... | 115             | 200          | 0.575                 | 0       | 0                |
| HALLMARK_UNFO...  | 59              | 113          | 0.522                 | 0       | 0                |

**Note:** An “NA” in the nominal and adjusted p-values of the results indicates that the test for that gene set was not conducted due to either not enough genes in a gene set or high expression spots. To enable testing of those failed



gene sets, lower the requirements by setting lower values of **Minimum number of genes** and **Minimum number of spots**.

The heatmap tab summarizes the adjusted p-values (FDR). The color scale presents non-significant p-values as red, and significant p-values as blue. In the context of the spatial gene set enrichment test, a sample (heatmap column) with evidence of a gene set (heatmap row) expression hotspot will appear as blue. Users can display sample-level metadata (“Sort samples by” dropdown, purple arrow). Users also have control over the number of gene sets to display (“Number of rows to show” slider, yellow arrow).

